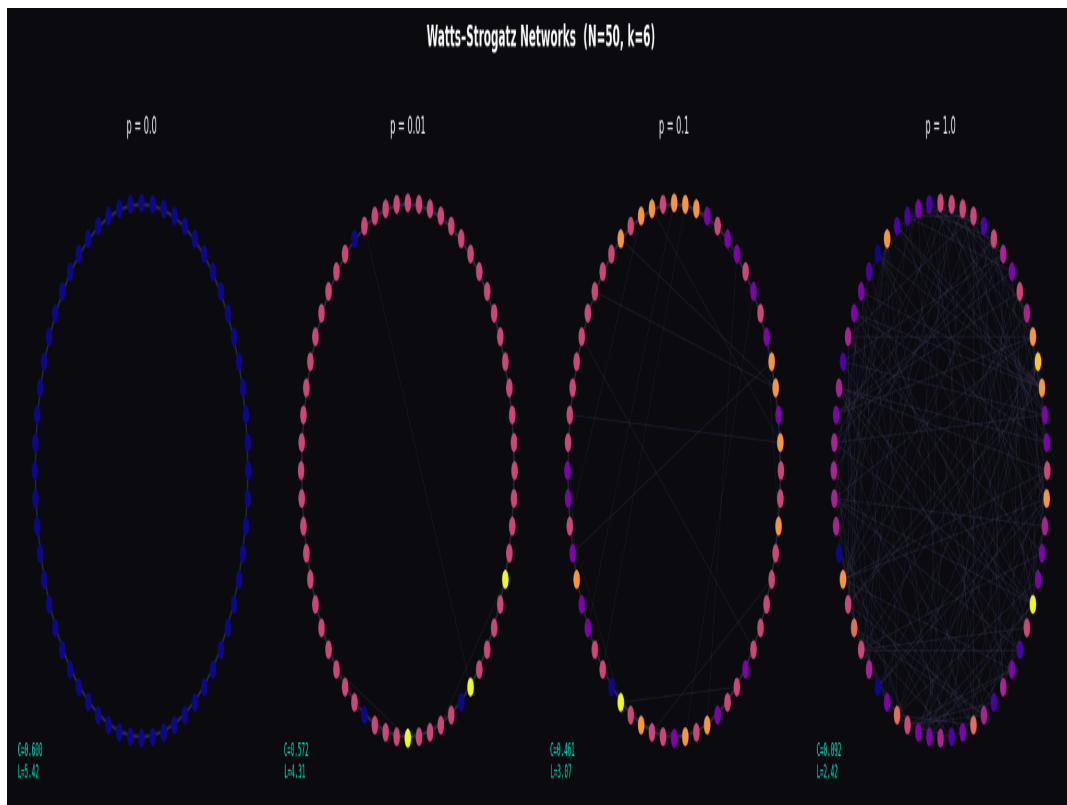
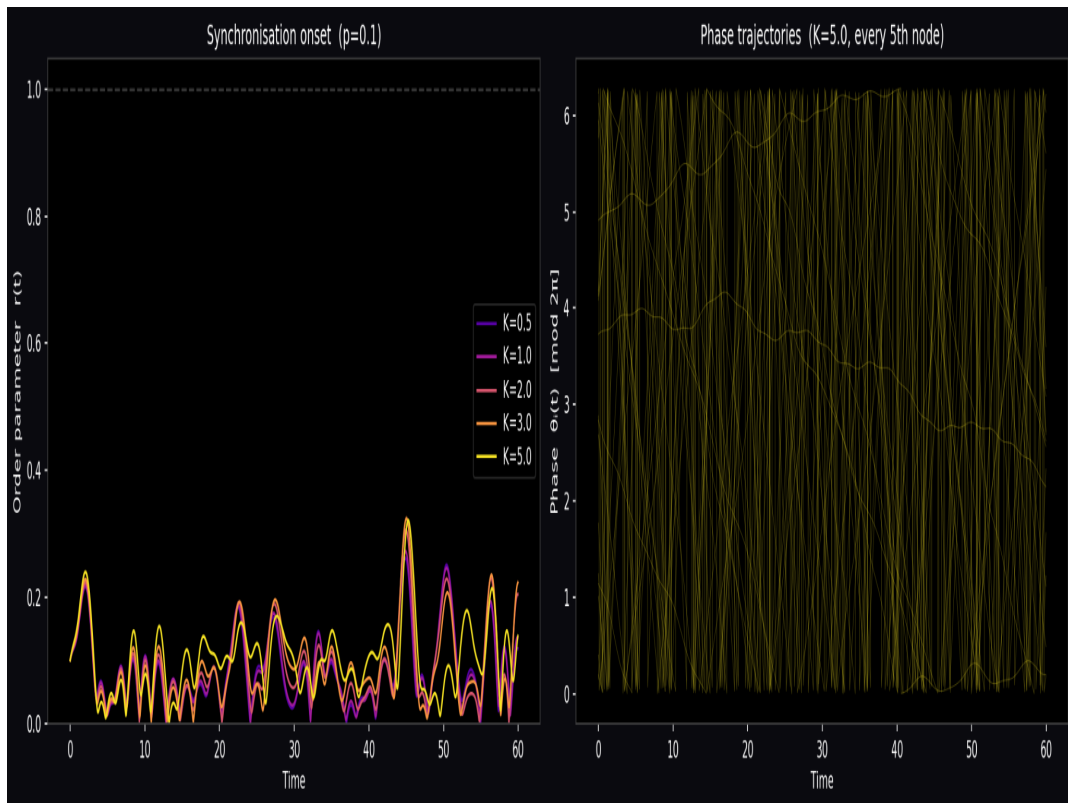


Module 4: Synchronisation on Watts–Strogatz Networks

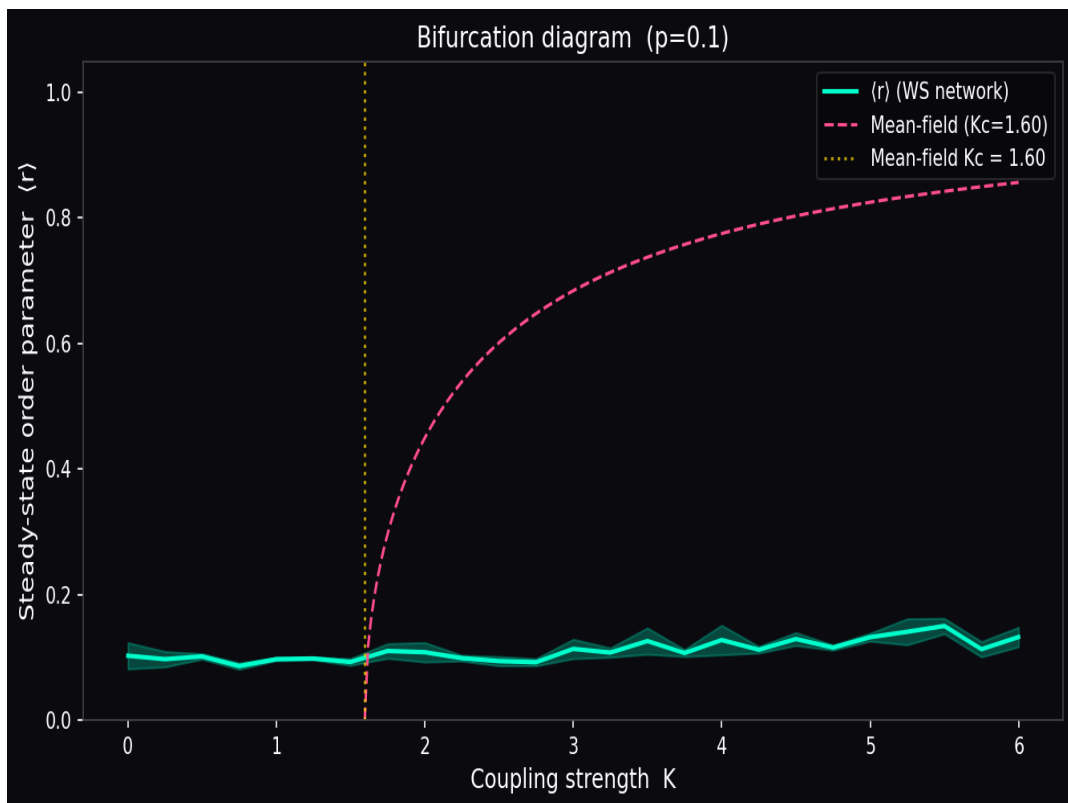
This module investigates collective synchronisation phenomena in Kuramoto oscillators coupled on Watts–Strogatz small-world networks. The figures summarise how network topology and coupling strength influence phase coherence, order parameter dynamics, and spectral properties.



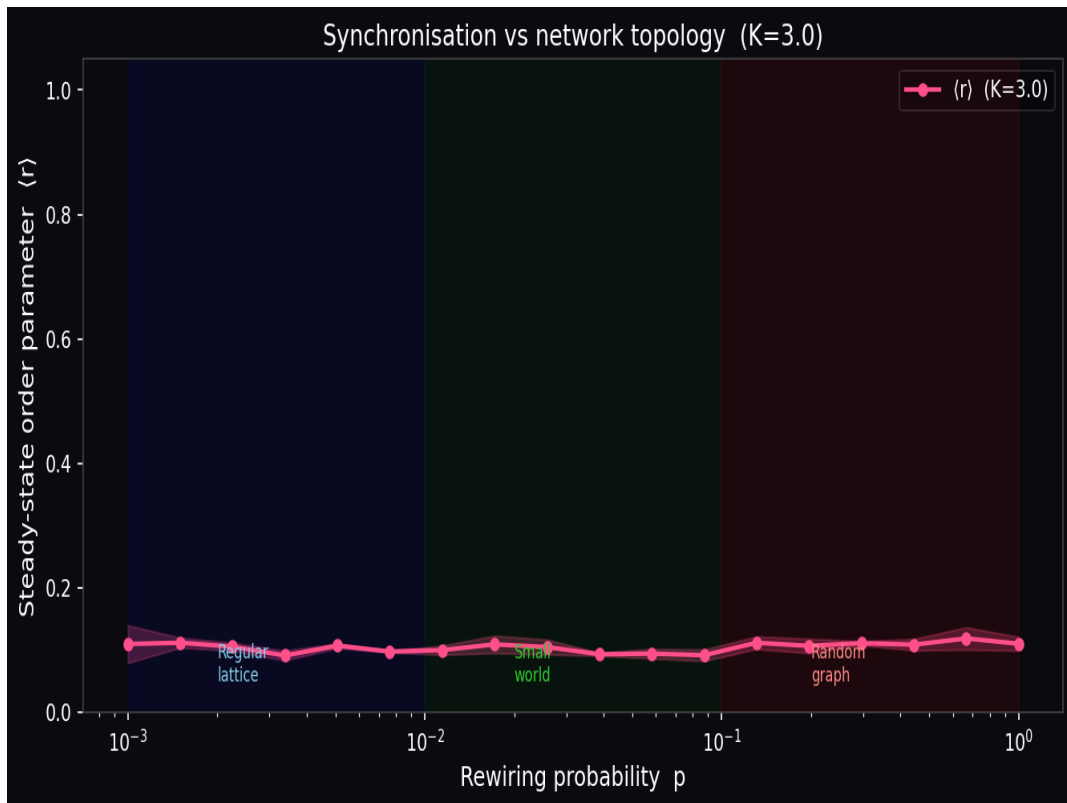
Watts–Strogatz networks for different rewiring probabilities p , illustrating the transition from regular lattices to random graphs.



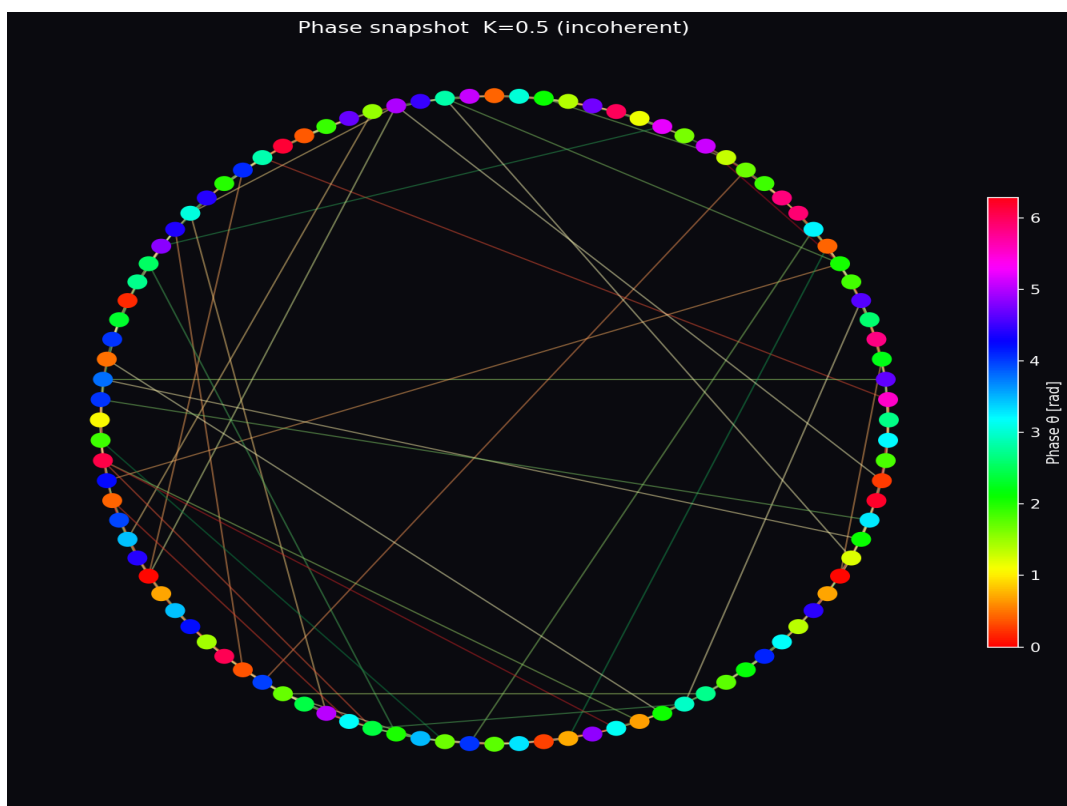
Temporal evolution of the Kuramoto order parameter $r(t)$ for increasing coupling strength, showing onset of synchronisation.



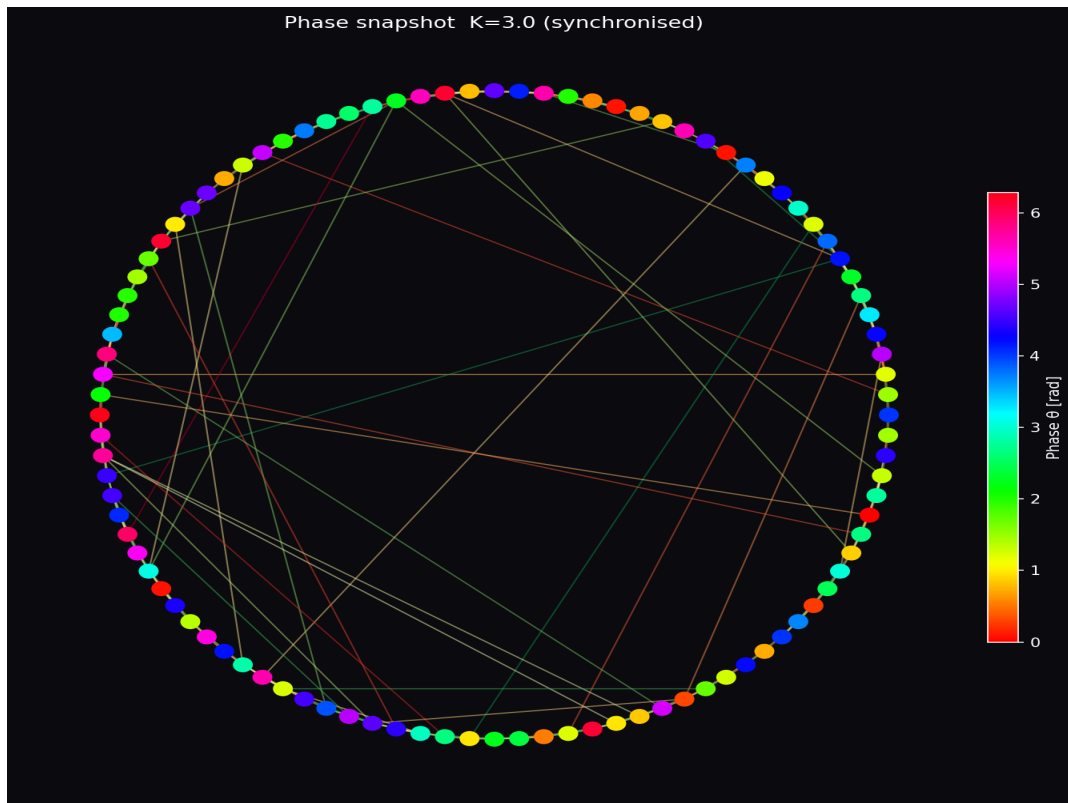
Bifurcation diagram of steady-state synchrony $\langle r \rangle$ as a function of coupling strength K .



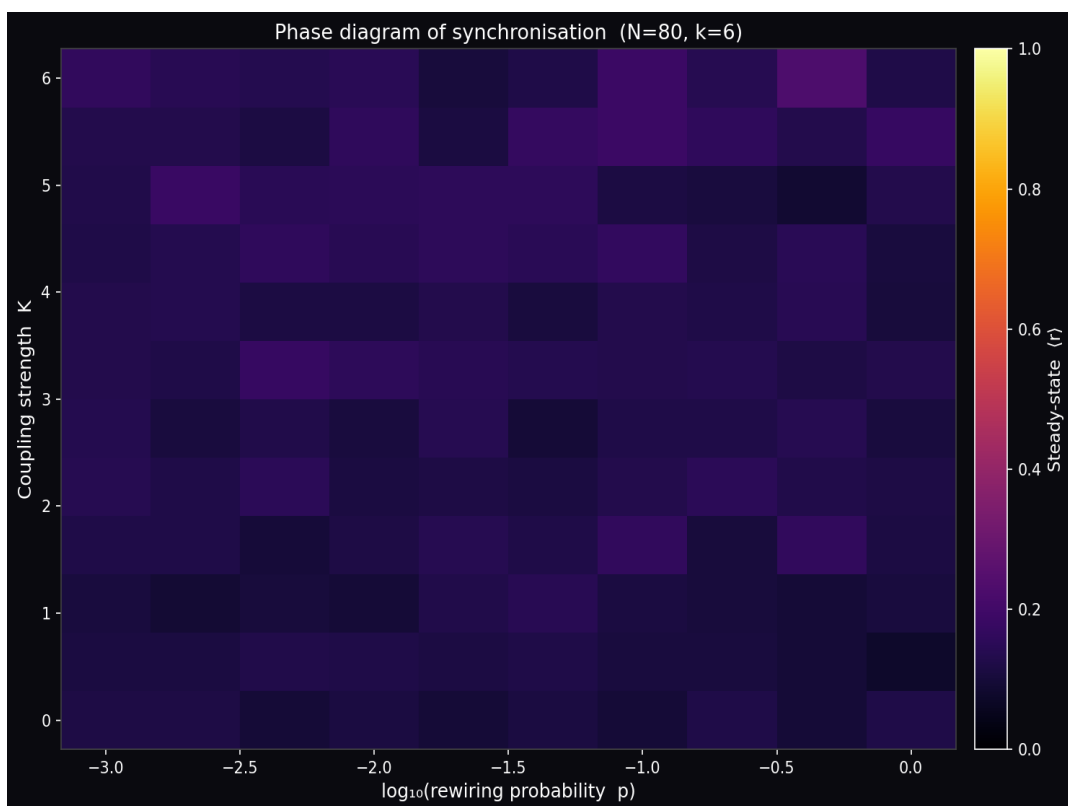
Dependence of synchronisation on network topology, quantified by rewiring probability p .



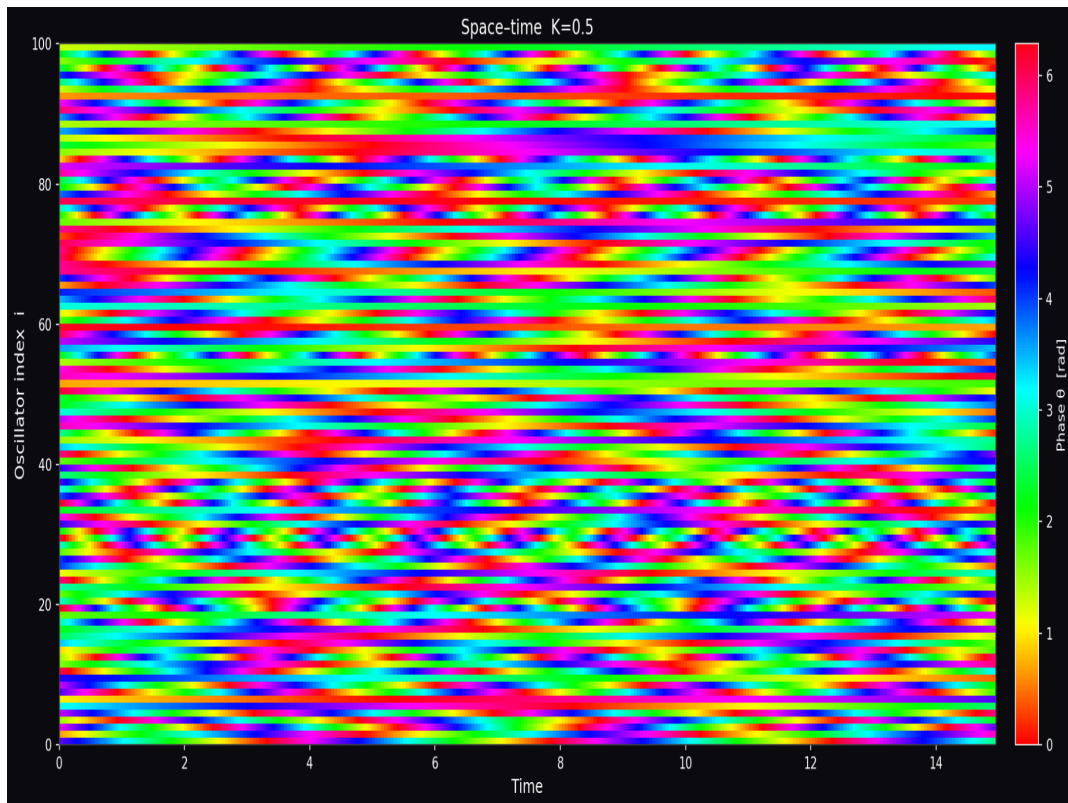
Phase snapshot at weak coupling (incoherent regime).



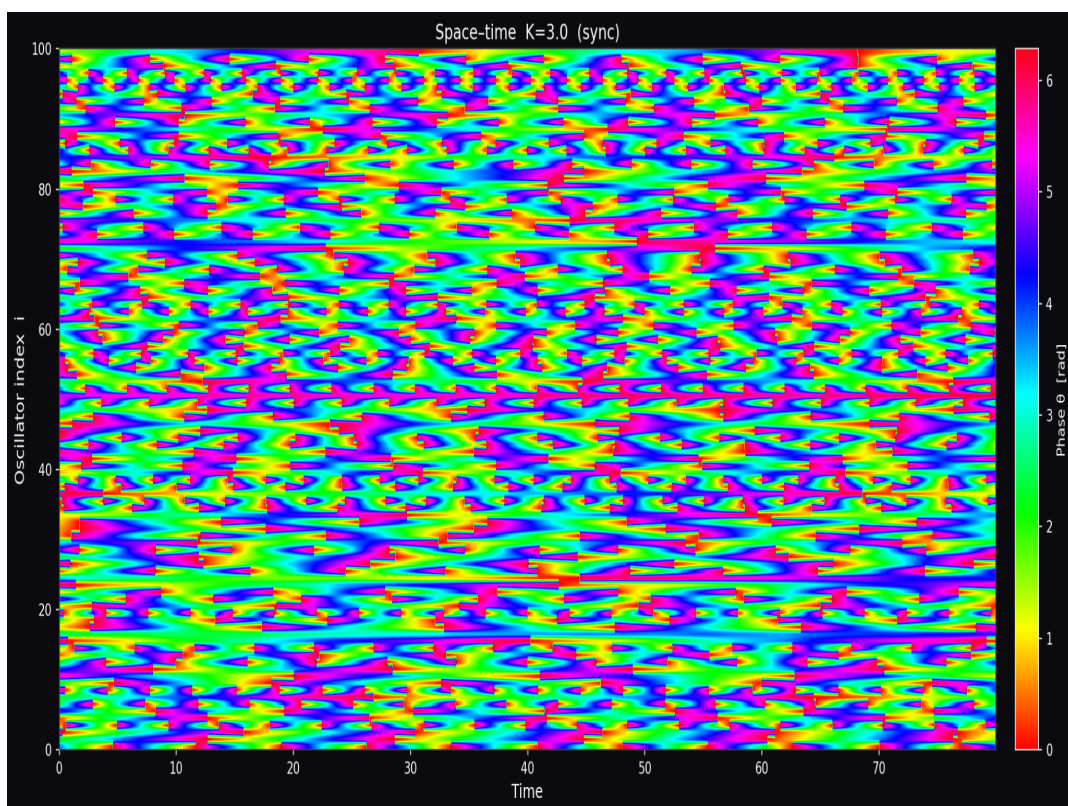
Phase snapshot at strong coupling (synchronised regime).



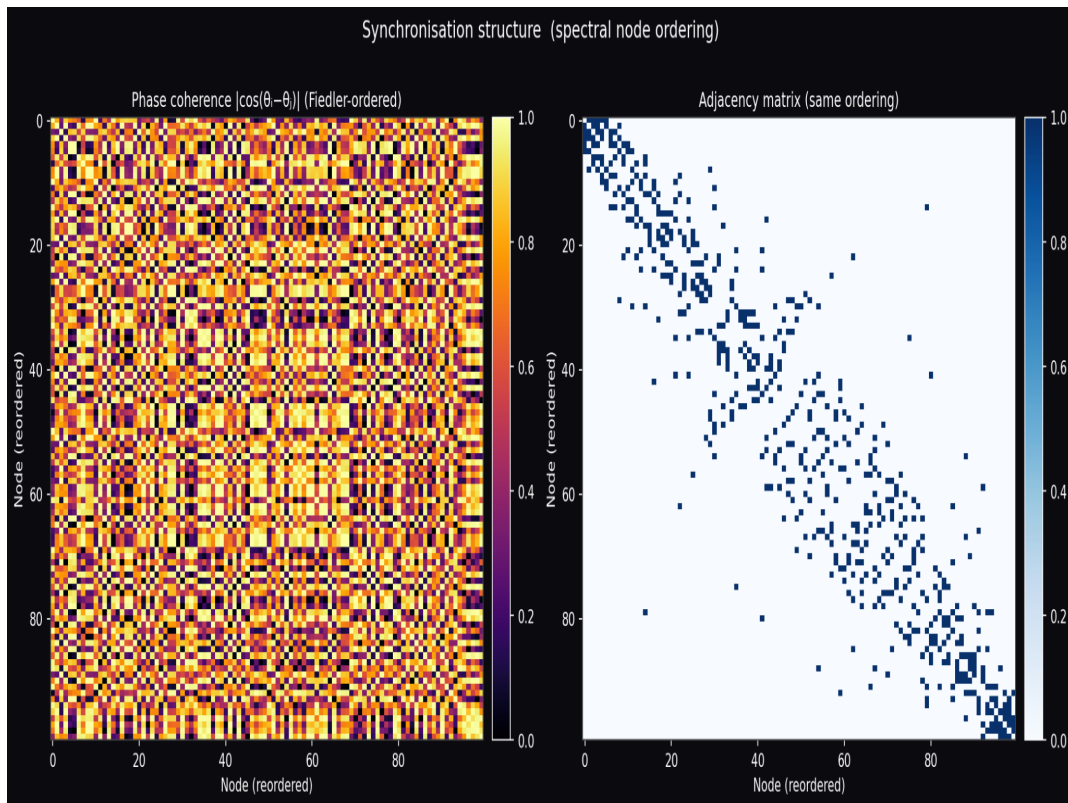
Phase diagram of synchronisation across coupling strength and rewiring probability.



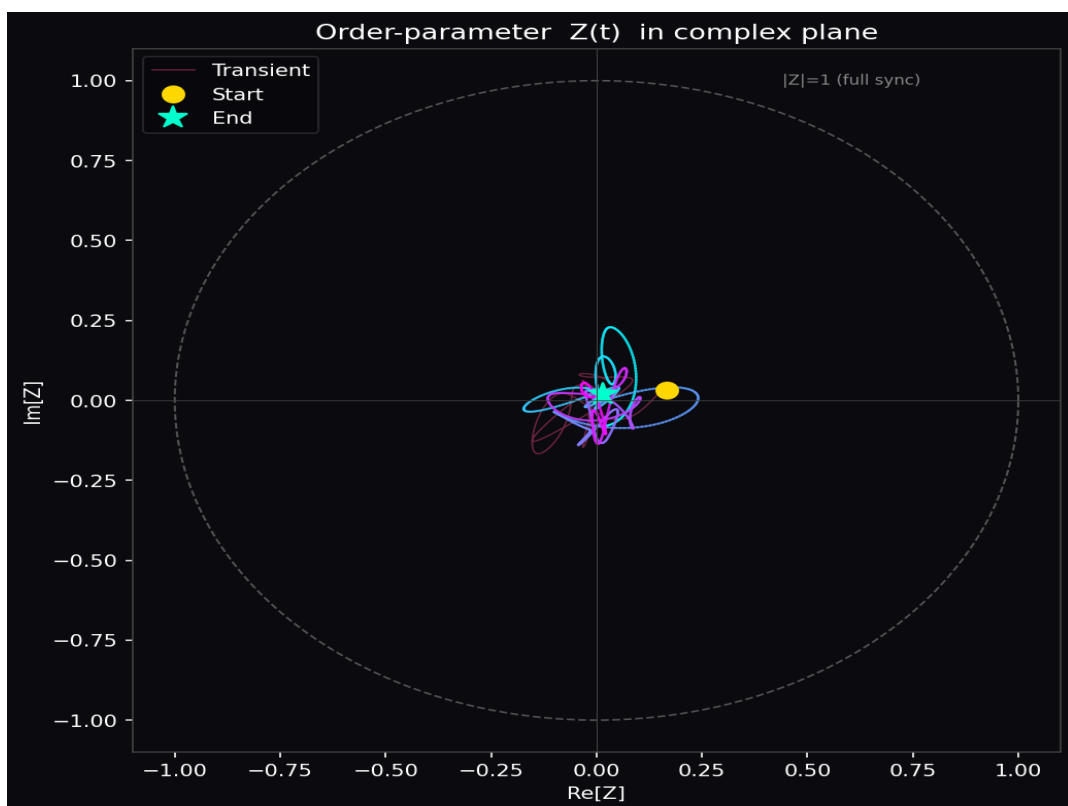
Spatio-temporal evolution of oscillator phases in the incoherent regime.



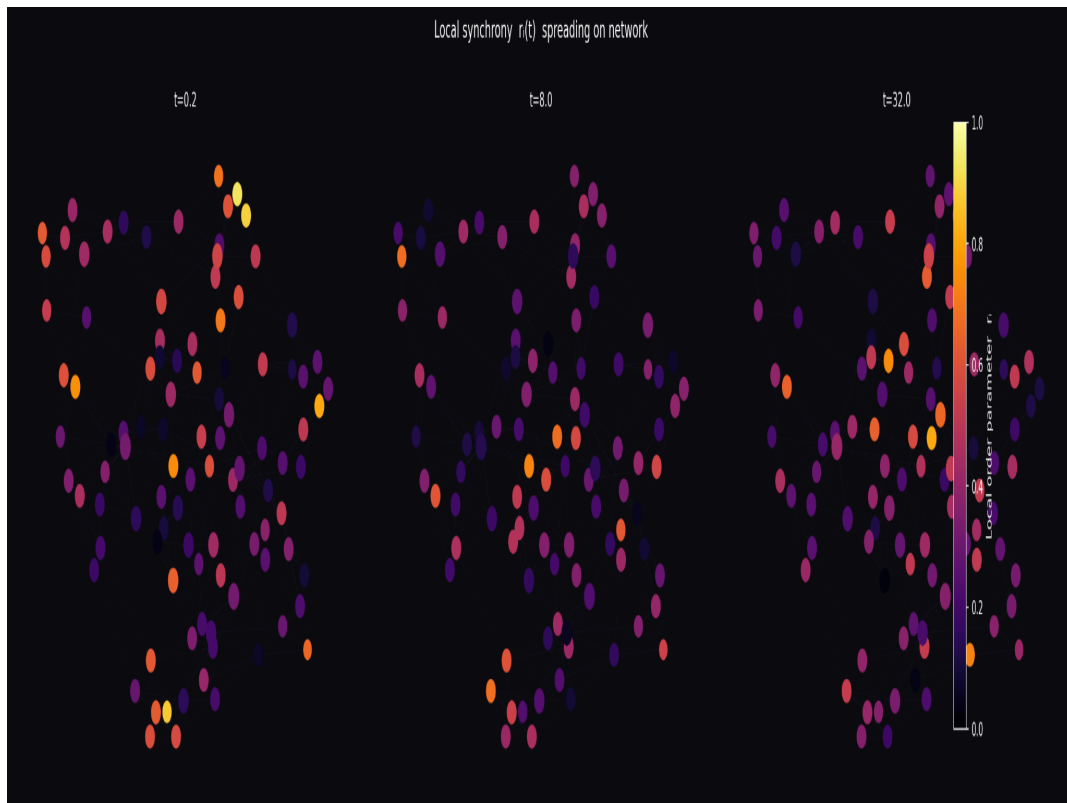
Spatio-temporal evolution of oscillator phases in the synchronised regime.



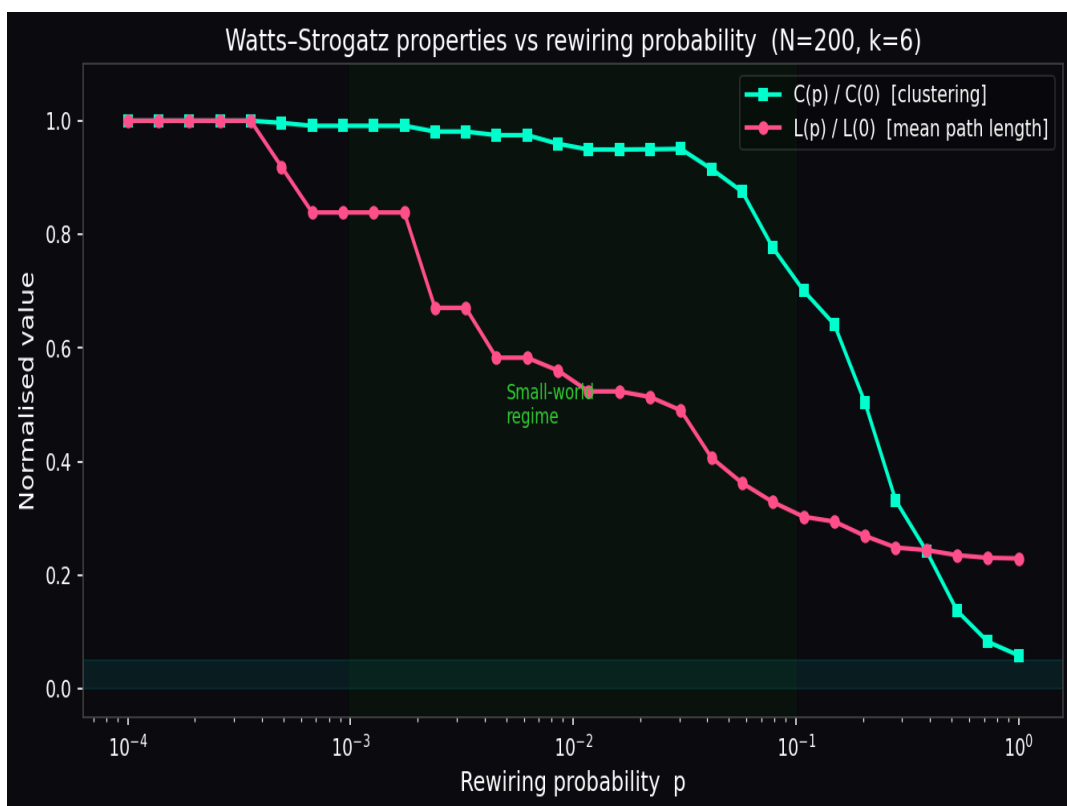
Phase coherence matrix and adjacency structure revealing synchronisation patterns.



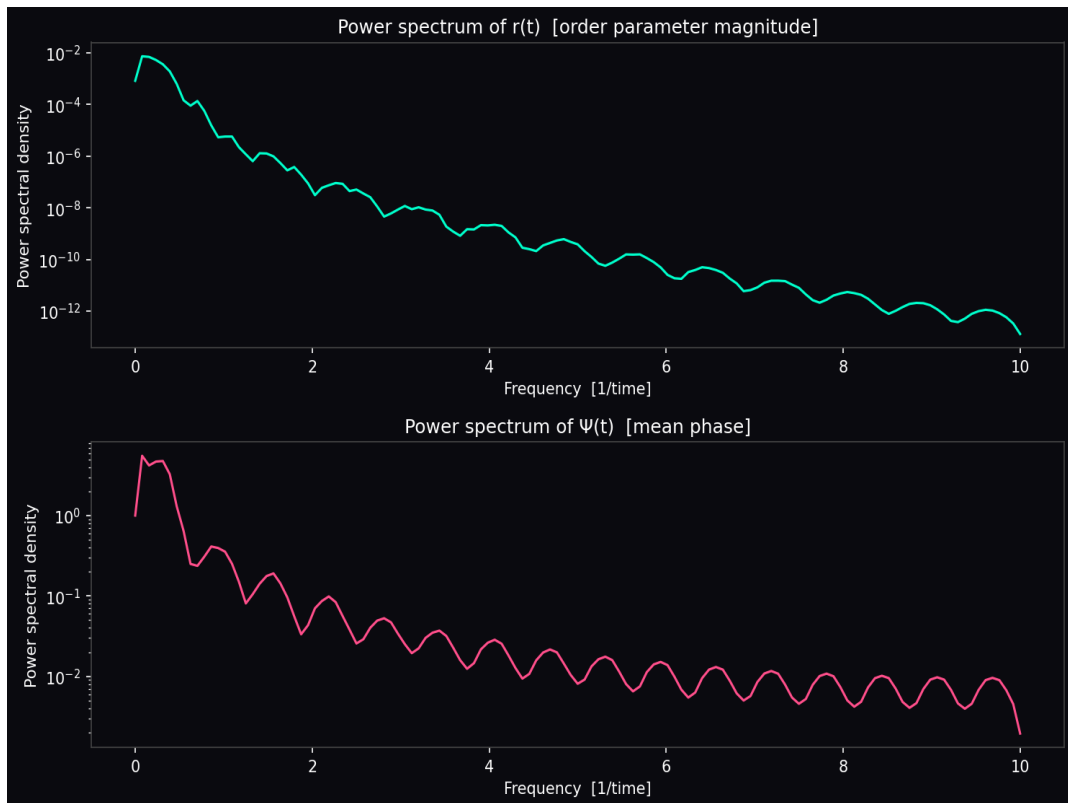
Trajectory of the complex order parameter $Z(t)$ in the complex plane.



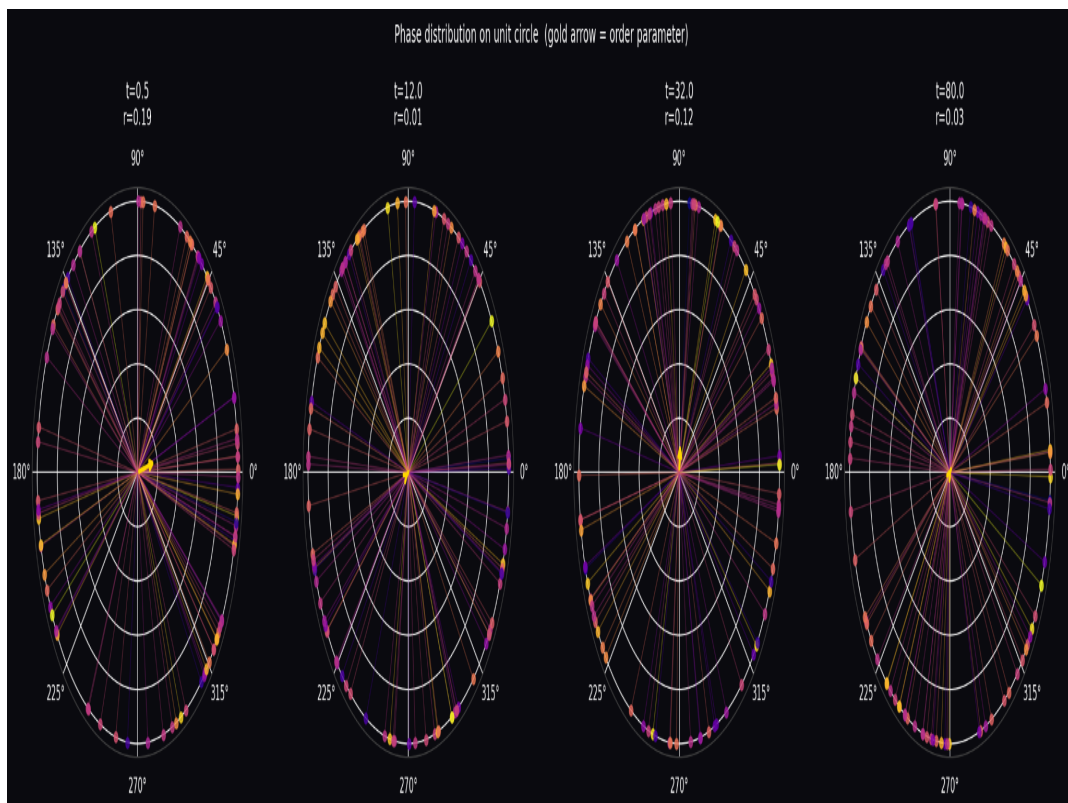
Spread of local synchrony $r_i(t)$ across the network over time.



Variation of clustering coefficient and path length with rewiring probability.



Power spectra of the global order parameter magnitude and mean phase.



Phase distributions on the unit circle highlighting collective coherence.

Reference

GitHub repository: <https://github.com/biswa-03/Modelling-Complex-Systems>

Jupyter Notebook: Part1_simulation_of_N_coupled_oscillators.ipynb